

MICHAEL B. GREENBAUM

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EDUCATION

Rowan University, Glassboro, NJ

Graduated May 2026

Bachelor of Science in Computer Science

Minor in Mathematics, Honors Concentration, GPA 3.890/4.0

Awards: Summa Cum Laude, CS Outstanding Senior Award, *YPIE* CS Honors Society, *PIIE* Math Honors Society, President's List for Semesters 1/2/3/5, Dean's List for Semesters 4/7, Rowan University Scholars Scholarship Awardee

Relevant Coursework: Object-Oriented Programming, Data Structures and Algorithms, Software Engineering, Computer Org., Database Systems, Calculus I–III, Discrete Math, Linear Algebra, Probability and Random Var., Statistical Analysis

EXPERIENCE

Digital Simulations Team, Lockheed Martin Corporation, Camden, NJ

January 2025–Present

Software Engineer; Associate Member Engineering Staff (E1)

Built a low-fidelity component factory that lets users quickly add components to the simulation with highly variable parameters. This low-fidelity, low-effort factory was intentionally designed to contrast with the high-fidelity, high-effort main simulation. Using the factory, users can define and test original component magnitudes faster; this is especially useful for new proposals and proofs of concept. Components are defined in a flat file, which is then parsed and injected as templates. At runtime, templates are instantiated, and the workarounds are performed to activate them.

Created a message processor which enabled two-way communication between a system and the main simulation. Messages are stored in a queue where they have simulation effects such as time offsets applied to them. They are then converted into their other type and sent. Additionally, hard-coded messages can be specified in a flat file, then parsed and injected into the system at a defined time; this mimics the main simulation without running it.

Constructed an external interface that bridged the local system and an external component while minimizing coupling. The external component is heavily abstracted and encapsulated, allowing the system to utilize its functionality without needing to understand its inner workings. The external component is referenced, its functions are used to obtain the desired data, and this usage is then wrapped into new functions with a reduced profile.

High-Performance Data Mining Lab, Rowan University, Glassboro, NJ

October 2022–January 2025

Project Manager, "GPU-Based Graph Mining"; Research Assistant

Led a multi-year team of researchers to develop a GPU-accelerated algorithm that solved an NP-hard problem 3,992x faster. First, the team investigated the maximal γ -quasi-clique problem by reviewing cutting-edge work. Then, we developed a baseline CPU program to outline our approach. Next, we moved the algorithm's processing to the GPU; the main novelty was the development of new data structures and processes to enable this transition. This is the first work to ever address the problem on the GPU, given its high task complexity and massively imbalanced workloads. Finally, parallel and distributed CPU techniques were used to accelerate the algorithm. The massive speedup and the novel techniques that enabled it culminated in publication at a top conference.

ACTIVITIES

First Author of Published Research: Michael Greenbaum, Ke Fan, Wajid Manzoor, and Guimu Guo. "GPU-Accelerated Maximal Quasi-Clique Mining." Dec 8, 2025. In 2025 IEEE International Conference on Big Data (BigData), pp. 152-161.

International Collegiate Programming Competition: Mid-Atlantic USA 2024 Division 2, First Place

LeetCode: Global Rank: 1,796,088 (Top 11.97%), Contest Rank: 176,796 (Top 20.31%)

TECHNICAL SKILLS

Programming Languages: Python, Java, C/C++, CUDA C++, MPI, Pthreads, OpenMP, Makefile, Bash, Sed, Grep, Find

OS, IDE, and Applications: Windows 10/11, Linux/Unix, VSCode, Visual Studio, VIM, GitHub, Atlassian Suite, Git, GDB, GCC

Security Clearance: Secret